

Student Assignment Brief

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Assignment Information

Module Name:	Advanced Machine Learning
Module Code:	ST7085CEM
Assignment Title:	STW7085CEM Advanced Machine Learning March Intake 2026 Coursework CW (Regular)
Assignment Due:	21-June-2026, 05:00PM

Assignment Credit:	15
Word Count:	4500
Assignment Type:	Coursework : Individual Report
Grading:	Percentage Grade (Applied Core Assessment).

Assessment Overview

You will be provided with an overall grade between 0% and 100%. You have one opportunity to pass the assignment at or above 40%.

Important Notice

The work you submit for this assignment must be your **own independent work**. More information is available in the 'Assignment Task' section of this assignment brief.

Assessed Module Learning Outcomes

The Learning Outcomes for this module align with the marking criteria which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task. The following module learning outcomes are assessed in this task

Assignment Task

Task and Mark distribution:

This coursework consists of two tasks, and you should attempt both and submit one Word or PDF file (or similar) for each task. Each task is worth 50 marks, and the breakdown for each task is provided with each task. This coursework contributes 100% to your overall module mark.

Task 1: The Machine learning algorithms for solving real-world problems in Regression, Classification, modeling data, and text mining

Individual/Group (of size 2) Research Paper: **50%** of the module mark

Context

During this module, you learn about different advanced machine learning techniques, associated concepts, and applications. We explored the Gaussian process model, which is a computationally efficient method for regression, classification, optimization, etc. We have also covered the Bayesian networks as promising tools for modeling the data with complex dependency structure.

Finally, you have learned how to use Dirichlet latent processes for unsupervised learning applications, particularly text mining.

In this assignment, you will have to select an application related to regression, classification, modeling unstructured data, or text mining problems, and explore how best to apply machine learning algorithms to solve them. The selected application for each of the methods mentioned above should have the following features:

1. Gaussian Process Regression and Classification: The application selected for any of these two methods must consist of at least four input variables and a single output variable. You must also implement Gaussian process classification by appropriately defining a threshold on the output variable to create a binary or multiple classes first and then apply Gaussian process classification to the categorized output.
2. Bayesian network: If you are choosing an application for this method, this application must consist of at least eight random variables. The random variables could be all discrete or continuous, or hybrid.
3. There is no restriction on selecting the application to apply the Latent Dirichlet allocation model for topic modeling.

There are some potential projects listed below, which could be studied to get some ideas. However, I strongly recommend you come up with your own idea(s) by reviewing these projects and some other relevant and recent articles.

1. This dataset from the UCI repository is quite interesting. The task is to predict the depth in the body (effectively, the depth along the spine) given the properties of a two-dimensional "slice" of the body. The hard part about this problem is that it is actually the output causing the input rather than the other way around. I have not had luck designing a good regression method for this data. Can you do this?
2. Find a Bayesian interpretation of elastic net regularization and compare this method for regression against "standard" Bayesian regression (with a Gaussian prior) on a dataset of your choosing.
3. Probabilistic PCA using Gaussian processes is a Bayesian interpretation of the classical PCA algorithm for dimensionality reduction. Implement Gaussian Process-based PPCA in Python, R, or Matlab, and compare its performance with other methods (such as "standard" PCA) on a dataset of your choosing.
4. Bayesian optimization is a very important issue with a wide range of applications. However, this was not fully studied during lectures, but it can be easily implemented using the Gaussian Process. The Python code and some examples can be found [here](#)!
5. The squared exponential covariance is widely used for Gaussian process regression. It is probably used in 90+% of all GP publications. That said, it is widely believed to be "too smooth" for many real-world regression tasks. Compare the squared exponential covariance versus the Matérn covariance on

several datasets via Bayesian model selection. How often is the squared exponential the right choice?

6. Latent Dirichlet allocation (LDA) is a Bayesian method for creating "topic models" of text documents. There are plenty of interesting text datasets available (e.g., DBpedia could be a good resource!). One idea would be to compare the behavior of LDA with other techniques, such as latent semantic analysis.

You may be able to get relevant datasets and ideas by visiting the following sites:

1. This competition site consists of some relevant data, and the relevant ideas could be developed by analyzing this data. Check also the dataset for Kaggle competitions.
2. This website has a fantastic compilation of 100 interesting, relevant datasets from all sorts of application areas.
3. The creators of libSVM have also compiled a great list of datasets, all in a standardized format. The libSVM codebase also includes libsvmread for reading these in MATLAB.
4. The UCI Machine Learning Repository is a mainstay in machine learning research. There is a wide range of datasets from many different application areas and with many different properties (large, small, high-dimensional, low-dimensional, classification, regression, etc.).
5. DBpedia is an amazing resource that automatically extracts structured data from Wikipedia. They have all sorts of data available for download in convenient formats. This tool can be used to extract labeled graphs from DBpedia, but there is so much more you can do.

The purpose of the first TASK of this coursework is to

- Examine the fundamental concepts of machine learning, their implementation, and application.
- Perform appropriate preparation of a dataset and evaluate the performance of different learning algorithms on this dataset.
- Gain practical experience in selecting machine learning algorithms for solving a real-life regression, classification, modeling data with complex dependency structure, or text mining problem.
- Demonstrate effectiveness in project teamwork and leadership.

You will be required to:

Ideally work in groups of 2 or 3, developing a paper/report by providing the details of contributions of each group member in developing/writing the project report. If you are working in a group of two, you need to consider developing at least one common

methodology and one individual technique for your analysis. For individual groups, at least two different analysis methods must be applied and compared.

- Actively participate in all activities;
- Consult with your tutor about your project work if needed during the support sessions. • You will write a proposal (maximum of 1 A4 page), giving the title of the project, the names of all group members, the description of the problem, and the plan of the work. You will need to submit this proposal to your tutor by Date. In case of any required changes, you will receive feedback from your tutor.
- Your final submission on TASK 1 will include a scientific paper (in 6 pages of A4, up to 4000 words), written individually or as a group based on the experience and the results gained during the group work. You will have to acknowledge the contributions of all group members in your paper by clearly stating the details of your individual contributions in a separate section.

You are encouraged to target a certain conference or journal and submit the proposed paper to it. You can either use the Machine Learning Journals template or any other single-column format from other relevant journals or conference sites.

The paper should broadly include the following sections:

- Abstract
- Introduction (where you introduce the problem along with a short literature review of related work; if the literature review is longer, it is recommended to be a section on its own).
- Problem and Data set(s) description (where you describe in detail the problem you want to solve and its significance)
- Methods (where you shortly describe the machine learning methods and/or other methods employed to solve the problem)
- Experimental setup (including data pre-processing, feature selection and extraction)
- Results
- Social, ethical, legal and professional considerations
- Discussion and Conclusions
- References

These are generic section titles, which you may adapt appropriately to the application/problem that is being investigated. You may include sections describing modifications of algorithms or developments that are novel and specific to your work. You may include figures, tables, pseudo code, and appendices with the actual code that has been developed.

The project's general guidelines and milestones:

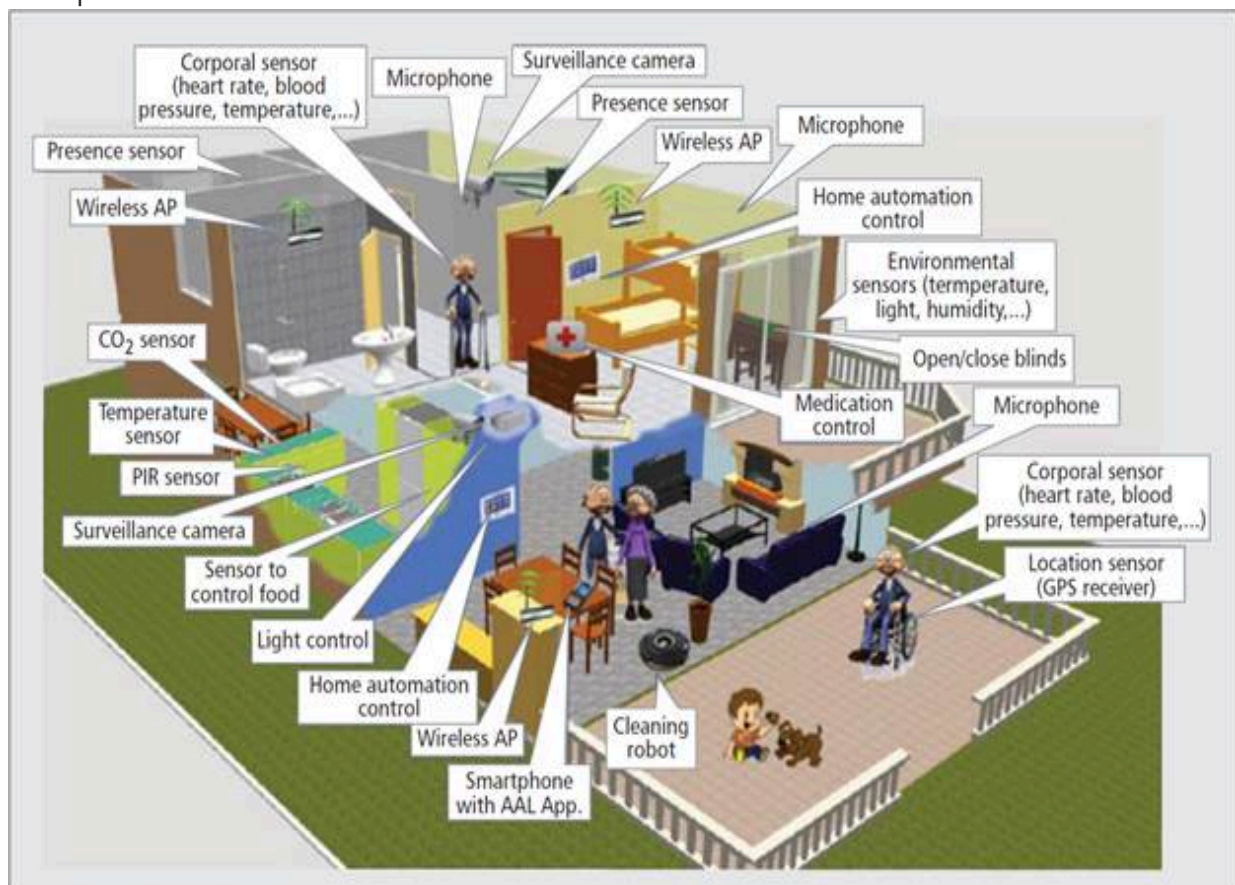
Please note, the following guidelines are good practice and should lead to better results, but you have the freedom to pick whatever is suitable for your style:

Working in groups of maximum 2 or 3, you must select a challenging real-world problem and one (or more) appropriate data set(s) as suggested above. You could also use the following links, which have numerous problems and data sets:

1. Learning Repository: <http://archive.ics.uci.edu/ml/>;
2. Kaggle competitions: <http://www.kaggle.com/competitions>;

Task 2: Evolutionary and Fuzzy Systems: Fuzzy Logic Optimized Controller (FLC) for an Intelligent Assistive Care Environment

Design and implement an FLC for controlling the environmental parameters of an intelligent flat for disabled residents, where the system needs to automate the regulation of environmental conditions and user preferences or the operation of assistive equipment, ramps, auto-adjusting furniture, kitchen worktops, HAVC, lighting, or water temperature control. The environment could be based on a room of choice in a small apartment. A more ambitious project might consider the aspects of the whole flat, but this is left to your choice. The environmental parameters to be controlled could be ambient temperature, thermal conformity, and lighting using actuators such as cooling fans, heaters/boilers, blinds, and dimmer switches. You might also consider other parameters such as TV or music volume control and power-down options for electronic devices and heating. Environmental parameters could be controlled based on monitoring sensors such as temperature, humidity, weather conditions, light levels, time of day, and level of activity/motion of the user, as well as mood and qualitative indicators such as user preferences.



FLC Design

The FLC should be based on determining the inputs and outputs of the system.

Depending on what control behavior(s) you decide the FLC should implement.

Note that, depending on the control behaviors you wish to implement, you can select to use a subset of the input sensors for example, so first think about the behavior(s) the FLC should control.

Design choices should be made to consider the type and number of fuzzy sets for the inputs and/or outputs of the FLC. A set of suitable control rules should be defined, which can be experimented with to achieve a good control performance of the chosen behavior(s).

The FLC should therefore implement the followings:

- Consideration of which Fuzzy Inference model to use: Mamdani or Sugeno (TSK) fuzzy models.
- Mapping the crisp input and output data into the designed fuzzy sets.
- Map input fuzzy sets into output fuzzy sets (for Mamdani model) based on a set of designed rules that capture the desired control behavior of the system.
- Employ appropriate inference operation (rule implication) that handles the way in which rules are activated and combined (composition and aggregation).
- The outputs of the fuzzy inference engine will define a modified output fuzzy set (for Mamdani model) that specifies a probability distribution of the control actions in relation to activated rules.
- Use an appropriate defuzzifier to convert the modified fuzzy outputs into non fuzzy (crisp) control values that can then be used to set up the actuation outputs.

Part 1 – Design and Implementation of the FLC (30 Marks)

Design and implement a demonstrable FLC, which can be a simulated system programmed in Matlab, FuzzyLite or Juzzy, see links below:

Matlab Fuzzy Logic Toolbox: <http://uk.mathworks.com/videos/getting-started-with-fuzzy-logic-toolboxpart-1-68764.html>.

<http://www-rohan.sdsu.edu/doc/matlab/toolbox/fuzzy/fuzzyt10.html>, Fuzzylite

<http://www.fuzzylite.com>, Juzzy <http://juzzy.wagnerweb.net>

- Provide suitable evidence of your implementation in the form of diagrams and screenshots of the different components. **(16 marks)**
- Discuss and justify your design decisions for the choice of fuzzy sets - membership functions, fuzzy rules, FLC inference mechanism selected, and defuzzification method that was chosen. Back up your explanations with evidence in the form of appropriate diagrams and screenshots. **(7 marks)**
- Perform analysis of the output behavior of the controller showing the rules of activation, controller output and control surface plots, demonstrating how the controller achieves the specified behaviors in relation to an operational scenario. **(7 marks)**

Part 2 – Optimize the FLC developed for Part 1 (10 Marks)

Consider the Fuzzy Logic Controller (FLC) for controlling the smart home you have designed for the above Part 1. The purpose of this part is to optimize the fuzzy controller you have previously developed. A data set of n input-output examples, (x_i, y_i) , $i = 1, 2, \dots, n$, is available to evaluate the performance of your controller.

Keeping the same structure of the FLC as you have used in Part 1, design a genetic algorithm to adjust the membership functions of the input and output variables of the FLC to optimize the performance of your FLC. Give details of the genetic algorithm you have used, i.e., problem encoding, genetic operators, fitness function. Some of you may have designed the FLC as a Mamdani model, while others may have used Sugeno models. Clarify what the length of the chromosomes used in your solution is. In case you have used a Mamdani model to implement your FLC, describe how the genetic algorithm solution would change if you were to use a Sugeno model for your FLC. Conversely, if you have used a Sugeno model in Part 1, describe how the genetic algorithm solution would change if you were to use a Mamdani model for your FLC.

Part 3 – Compare different optimization techniques on CEC'2005 functions (10 Marks)

Choose two functions from the CEC'2005 suite of benchmark functions available here:

<http://www.cmap.polytechnique.fr/~nikolaus.hansen/Tech-Report-May-30-05.pdf>

More details about the special session at CEC'2005 can be found here:

<https://www.alroomi.org/benchmarks/cec-database/cec-2005>

Some of the links in these pages are broken, but you will be able to download the Matlab code for the functions if you click "Resources Database (Different Formats) [Download]" on the last web page indicated above. This part is to compare the performance of at least 2 different optimization techniques on the two functions you have chosen, for both $D=2$ and $D=10$, where D is the number of dimensions. If you want to challenge yourself, you may try higher dimensional spaces, for example $D=100$, but this is optional. As optimization techniques to compare in this part, you may choose Genetic Algorithms, Particle Swarm Optimization, Simulated Annealing or other optimization methods available in the Global Optimization Toolbox or the Optimization Toolbox in Matlab or developed as standalone programs by yourself. To make the comparison meaningful you would have to run each optimization algorithm 15 times and report the average performance (including the standard deviation of the obtained results), as well as the best and the worst performance among the 15 runs. You may try to compare your results with results reported in the literature on the same functions. In your report, you should include the description of the functions you have selected, the Matlab code for those functions, the results obtained, and the parameters of the optimization algorithms used to obtain the reported results, any other Matlab scripts or code used in your simulations, convergence graphs, etc.

Submission Instructions

Requirement	Details
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File Naming	NAME_studentID
File Format	.docx/.pdf format
Submission Method	Campus 4.0 platform (submission link provided 2 weeks before deadline)

Marking and Feedback

How will my assignment be marked?

Your assignment will be marked by the Module Team using standardized criteria.

How will I receive grades and feedback?

Provisional marks will be released once internally moderated. Feedback will be provided alongside grades release within 2 weeks (10 working days).

What will I be marked against?

Details of the marking criteria for this task can be found in the Assessment Marking Criteria section at the end of this brief.

Grade Requirements

You must achieve 40% or above to pass this assessment. Ensure you understand the marking criteria for successful completion.

Assignment Support and Academic Integrity

Getting Help

If you have any questions about this assignment, please meet your respective module leader or teacher for more information.

Language Standards

You are expected to use effective, accurate, and appropriate language within this assessment task.

Academic Integrity

The work you submit must be your own. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work, **excluding Artificial Intelligence (AI)**. We use detection software and make routine checks for evidence of academic misconduct. Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found in Student

handbook in Campus 4.0. All cases of suspected academic misconduct are referred to for investigation, the outcomes of which can have profound consequences to your studies.

Support for Students with Disabilities

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms, contact the Student Support Office for assistance.

Unable to Submit on Time?

If events prevent you from submitting on time, guidance on extenuating circumstances is available in the Student Handbook or from the Student Support Office.

Administration of Assessment

Module Leader Name:	Shrawan Thakur
Module Leader Email:	stw0049@softwarica.edu.np
Assignment Category:	Proposal
Attempt Type:	regular
Component Code:	cw

Assessment Marking Criteria

Marking Scheme for Task 1 Mark:

1. Proposal • The description of the problem and the initial plan of the work; • The initial and correct application of the proposed methods for the selected dataset; <i>Notes: A. You will not get the full marks for this section if you submit your proposal late.</i> <i>B. If the final submission of your CW is different from what you propose in your proposal, you will not get any marks for parts 2 & 3.</i>	5
2. Technical quality 1. Rigor and extent of the experiments. 2. Correct application of the selected algorithms and suitability of the methods. 3. Data preparation: technical quality. 4. Extent of evidence of running the experiments provided in appendices.	5 5 5 4
3 Evaluation 1. Evaluation and discussion of the results. Why are the results important? How would the results be useful to other researchers or practitioners? 2. Is this a “real” problem or a small “toy” problem? How does the paper advance the state of the art? Notes about valid codes: • All your programming code should be included in the Appendix of your report or provided via a valid GitHub link. Please display them in a structured way (put headings for each task implemented in your CW), with appropriate comments/annotations. • You need to attach the original R code (or Python/Matlab), NOT the screenshots of the code. • The code will be marked as part of the above marking scheme (for all the tasks in this coursework, you will need to provide the corresponding code; when you describe/discuss the tasks in the main text of the report, please reference the corresponding code section in the Appendix or link).	7 2
4. Social, ethical, legal, and professional considerations related to the problem in question.	2
5. Clarity of the writing: 1. Is there sufficient information for the reader to reproduce the results? Is the language used in the paper good? 2. References and general presentation; Are results clearly presented, with appropriate visualizations?	5 4
6. Originality: 1. Is there some original approach to the problem or original use of techniques?	3 3

2. Is there any (and how much) difference from previous contributions? presentation:	
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Parts and Mark distribution for Task 2:

Part 1	30 Marks
Part 2	10 Marks
Part 3	10 Marks

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